

Worksheet 6 *Exponential growth to equilibrium*

Prepares you for lab manual pages (TTM plan for L2 S6 is blank) (section L2 S6)

Name:

Name of TA:

1. Suppose F satisfies the change equation $dF/dt = 0.3F$ and the initial condition $F(0) = 100$.

- (a) Sketch a time series for F on a *semilog plot*. You may choose your logarithm base (2, 10, or e); label your vertical axis accordingly.



- (b) If $F(t) = Ce^{kt}$, then what is k and what is C ? $k =$ $C =$

- (c) What is $F(1)$? Round to the nearest whole number.

- (d) How much time does it take for F to double?

2. Write down three different functions that satisfy the change equation $F' = 2F$.

3. Suppose that P satisfies the change equation $P' = 2 - 0.2P$.

- (a) What value of P is the equilibrium point?

- (b) Draw two time-series plots below for the state variable P : one starting at $P = 1$ when $t = 0$, the other starting at $P = 20$.

